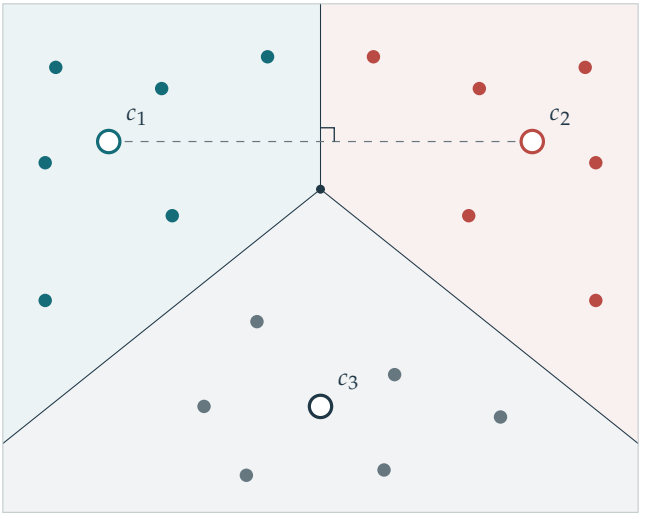

$$V_j = \{x : \|x - c_j\| \leq \|x - c_\ell\| \text{ for every } \ell\}$$

Dashed segment: two centroids. Its perpendicular bisector is their common cell edge.